



STINK3024 Machine Learning
Session II 2025/2026 (A252)
Assignment 1 Descriptive Analytics (10%)
LOC 3b

Type: Group work (4 members per group – as determined)

Release date: 9 April 2026_ Week 4

Submission: 23 April 2026_ Week 6

Assignment Goals

By the end of this assignment, students should be able to:

1. Identify and collect relevant datasets from reliable primary or secondary sources suitable for descriptive analytics.
2. Recognize proper data preprocessing techniques such as cleaning, transformation, and integration to ensure data quality and consistency.
3. Perform descriptive statistical analysis (e.g., mean, median, mode, standard deviation, frequency distributions) to summarize key data characteristics.
4. Visualize data effectively using appropriate graphs, charts, and dashboards to uncover patterns and trends.

Term and Condition: A penalty of 2 marks will be applied for plagiarism or late submission

Objective: To expose student to the process of data collection extracting and descriptive analytics

1. Introduction

Clearly define the project topic, objective, and motivation. Explain:

- **Problem Statement:** What issue is being addressed?
- **Purpose of Analysis:** Why is it important to analyze this dataset?
- **Context:** Brief overview of the industry or domain.
- **Goals:** What insights or outcomes are expected from the analysis?

Example: This project aims to analyze sentiment in Twitter data to understand public opinion trends around climate change policies.

2. Related Works

Provide a brief literature review summarizing existing research or projects:

- **Previous Studies:** Mention 15–20 academic papers.
- **Methods Used:** Highlight techniques, algorithms, or data processing approaches.
- **Gaps Identified:** Indicate what existing work lacks or what your project improves upon.

Example: Past research has applied TF-IDF and logistic regression to classify sentiment in tweets (Go et al., 2009), but few studies have focused on real-time public opinion tracking using transformer-based models.

3. Methodology

Detail the step-by-step approach followed:

3.1 Data Collection

- Source of the data (APIs, public repositories, surveys, etc.).
- Rationale for selecting the dataset.
- Dataset size and structure.

3.2 Data Preprocessing

- Handling missing data, duplicates, and noisy entries.
- Tokenization, lowercasing, stemming/lemmatization (for text).
- Normalization and encoding techniques for categorical variables.

3.3 Feature Engineering

- Describe features created or selected from the dataset.
- Explain relevance of each feature to the problem.

3.4 Vector Representation

- Describe how raw data is transformed into numerical vectors:
 - **For Text Data:** Bag of Words, TF-IDF, Word2Vec, GloVe, BERT.
 - **For Categorical Data:** One-hot encoding, label encoding.
 - **Dimensionality Reduction:** PCA, t-SNE (if used).

Include flow diagrams or preprocessing pipelines where appropriate.

4. Data Descriptive Analysis

Conduct exploratory data analysis to understand dataset characteristics:

4.1 Summary Statistics

- Mean, median, standard deviation, skewness, etc.

4.2 Visualizations

- Histograms, bar charts, pie charts, box plots, scatter plots.
- Trend lines or time-series plots if data is temporal.

4.3 Correlation Analysis

- Heatmap or correlation matrix to identify strong relationships.
- Multicollinearity issues and steps taken to address them.

4.4 Initial Insights

- Highlight patterns, anomalies, or hypotheses for further analysis.
- Mention how these insights guide future modeling or experimentation.

5. References

Students should use APA or IEEE style for referencing. Here are some example references to start with:

Sample References (APA Style):

1. Go, A., Bhayani, R., & Huang, L. (2009). *Twitter sentiment classification using distant supervision*. CS224N Project Report, Stanford, 1(12), 2009.
2. Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). *Efficient estimation of word representations in vector space*. arXiv preprint arXiv:1301.3781.
3. Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of deep bidirectional transformers for language understanding*. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 4171–4186.
4. McKinney, W. (2010). *Data structures for statistical computing in Python*. In Proceedings of the 9th Python in Science Conference, 51–56.
5. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). *Scikit-learn: Machine learning in Python*. Journal of machine learning research, 12, 2825-2830.

Submission Requirements:

- Format: Word document
- Length: 15–20 pages including visuals and references.
- Include clearly labelled sections and code (if applicable).